

EXPERIMENT - 3

3. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.

Aim

To read an image using Python and OpenCV and detect the outline/edges of the image using the Canny edge detection function.

Procedure

1. Import the OpenCV library.
2. Read the input image using `cv2.imread()`.
3. Convert the image to grayscale using `cv2.cvtColor()`.
4. Apply the Canny edge detection function using `cv2.Canny()`.
5. Display the original image and the detected outline.
6. Wait for a key press and close the image windows.

Program

```
import cv2

image = cv2.imread("image.jpg")

if image is None:
    print("Image not found!")
else:
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    edges = cv2.Canny(gray, 100, 200)

    cv2.imshow("Original Image", image)
    cv2.imshow("Canny Edge Image", edges)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

Sample Input

image.jpg



Sample Output



Result

Thus, the image was successfully read and its outline was detected using the Canny edge detection function in Python and OpenCV.